Is Lotka–Volterra-type of model right for five-element network in TMC?

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October 4, 2026

Abstract

1 Introduction

Traditional Chinese medicine (TCM) has been widely recognized and becomes an invaluable component in alternative medicines in the west [**?**] though it is not always viewed correctly. Chinese medicine is the result of science over long time as it is the result of continuous critical thinking and extensive clinical observations and testing [**?**]. It is a coherent and independent system of thoughts and practice that have been developed over 2000 years. As Manfred Porkert [**?**] rightly pointed out, Chinese medicine is not less logical than the Western, just less analytical.

TCM is an attractive alternative for several reasons. First, TCM takes holistic approach, which takes into account of not only interactions among different components within the whole body, but also external environment such as season, time of the day, the place of living and so on. This approach is very different from that of modern medicine, which essentially treat a patient as an independent sick organ with a bunch of testing numbers. It is so often that disease spread from one organ to another. TCM has the principle that strengthen the still healthy organ that is directly impact by the ill organ before treat the sick one so to prevent spread. This can only be achieved by knowing the overall relations between the organs. Second, because of holistic approach, TCM natually uses multi-target approach instead of single-target as modern medicine usually does (add citation). This approach could provide a balanced and effective treatment with less side effect. Third, TCM focuses on health, not disease. For example, it is so common that we feel something is going wrong, for example feeling lack of energy, but all medical tests show nothing wrong. TCM focuses on identifying the root problems. As said in Huang Di Nei JIng, the best doctor is the one who bring patient back to true healthy before the patient even gets sick. Finally, TCM is very accessible to average people. In the past, Chinese has some general idea on diagonizing and even able to construct some treatments by themselves for minor problems.

Imagine that if everybody can identify themselves health issue before gets sick, then the immense cost in medical can be saved, the pollution related to drugs can be dramatically reduced, and people can live healthier and longer and happier.

TCM is developed based on theories and principles described in the ancient work Huang Di Nei Jing. The main theories are based on two networks: a five-node network called five-element system, and a two-node network called yin-yang relation. Diagonsis and treatments are conducted based on the dynamics of the two networks. In this work, we aims to introduce and understand the theories from network dynamics point of view.

Network motif was proposed to understand dynamics of complex network systems by looking at small sub-networks that occur more often than usual or functionally building blocks [**?**, **?**]. It is known that the dynamics of a sub-network in general may not reflect the dynamics of the whole system. However, if the sub-network is relatively independent, then its dynamics could be manifested in the dynamics of the whole system. For example, the sub-network is an isolated network, or it does not have input from remaining networks. Interestingly, in TCM, major principles in diagnosis and treatments are based on the sub-networks of the five-element network.

The five-element network have been proposed to be modeled by a Boolean network system [**?**], or a system of differential equations [**?**, 1, **?**, **?**]. We are interested in Lotka–Volterra-type of model that was proposed in [1]. Lin *et al.*[1] studied the dynamics of the network at several special sets of parameters and explore possibility of chaotic dynamics. Their work was carried out from math interest instead of the applications in TMC. To make a model meaningful, we need to make sure the model captures some key characteristics that are observed in applications. For example, a healthy system needs to be stable in the sense that perturb initial conditions, or perturb parameter values, the overall dynamics stay the same.

In this work, we study equilibrium states and their stability of . explore the overall stability as we assume that a healthy human being should have stable dynamics. We thus explore overall stability of the system Lin's model and a modified version of the model.

2 The five-element network and principles in practice

OUTLINE:

1. FIIVE - ELEMENT NETWORK
2. THE PRINCIPLES IN PRACTICE FOR TREATING ILLNESS

The five-element network consists of the five elements: wood, fire, earth, metal and water. The two types

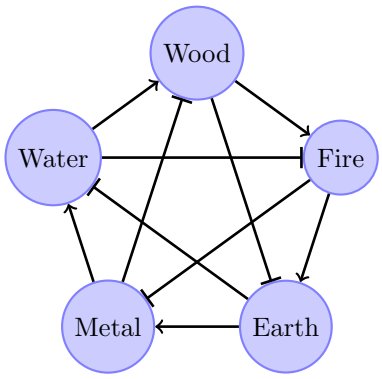


Figure 1: Five-element network and motifs approach in practice

of arrows in the fig. 1 represent two primary inter-element relationships: **Sheng (Generating $\rightarrow$) cycle.** and **Ke (Controlling, $\dashv$) cycle.**

Principles in practice. ADD THOSE DESCRIPTIONS

3 Dynamics of Lin's Model

Our starting question was whether the five-element network could maintain an active balanced state after one component was disturbed. A trajectory that happens to remain at equilibrium does not answer this question: the important test is whether nearby states return to that equilibrium. We therefore examined both local stability and the response to initial perturbations. This section introduces the Lin reference model and the baseline tests; Section 4 discusses the modified model and the broader parameter comparisons.

Throughout, "balance" means an equilibrium with equal positive activities at the five nodes. This is a convenient reference state for a model whose coefficients are identical across nodes. It is not an assertion that real organs have equal activities, or that mathematical stability alone defines health. The experiments below are preliminary comparisons at specified parameters, rather than calibrated physiological predictions.

3.1 The reference model and the meaning of balance

We order the nodes as Wood, Fire, Earth, Metal, and Water, and take indices modulo five. The predecessor $i-1$ is the mother, $i+1$ is the child, $i-2$ is the controller, and $i+2$ is the controlled target. In this ordering, the homogeneous model of Lin and Han [1] is

$$\dot{x}_i = ax_i - bx_i^2 + cx_ix_{i-1} - dx_ix_{i-2} - gx_ix_{i+1} - hx_ix_{i+2}. \tag{1}$$

The terms describe self-growth, self-limitation, nourishment received, restraint received, a cost associated with the child, and a cost associated with the controlled target. All five activities are state variables; "six-term" describes the equation, not the number of nodes. For positive coefficients, the positive balanced level is

$$s_{\mathrm{L}} = \frac{a}{b+d+g+h-c}, \qquad b+d+g+h-c>0. \tag{2}$$

Existence and stability are separate questions. Equation (2) identifies a possible balance; it does not tell us whether the network returns there after a disturbance.

3.2 Can the baseline balance resist a small disturbance?

We first used the reference choice $(a,b,c,d,g,h) = (1,1,1,1,1,1)$, which gives $s_{\mathrm{L}} = 1/3$. To test local stability, we evaluated the Jacobian at $s_{\mathrm{L}}\mathbf{1}$ and plotted its five eigenvalues. An eigenvalue with positive real part indicates a direction in which a small disturbance grows; all negative real parts imply that sufficiently small disturbances decay. Figure 2 also includes the modified-model baseline discussed in Section 4 so that the two behaviors can be compared directly.

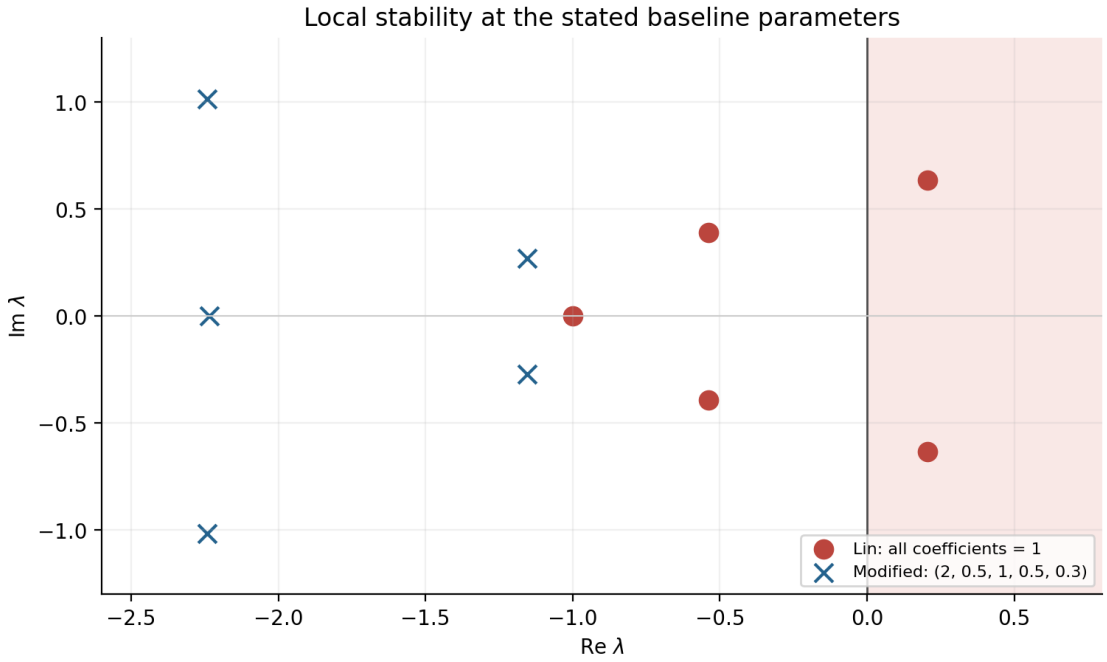


Figure 2: Local stability at two illustrative baselines. Lin uses $(a,b,c,d,g,h) = (1,1,1,1,1,1)$, with balanced level $1/3$. The modified model uses $(a,k,c,g,h) = (2,0.5,1,0.5,0.3)$, with balanced level approximately 2.670286. The vertical line separates negative and positive eigenvalue real parts. The baselines are specified examples, not a matched test of one changed mechanism.

Two Lin eigenvalues lie to the right of the imaginary axis, at approximately $0.206011 \pm 0.634038\,\mathrm{i}$. Thus the balanced state exists but is unstable. The nonzero imaginary parts indicate an oscillatory component in the growing linearized disturbance, rather than a purely monotone departure. This gives a concrete expectation for the time-domain experiment: a generic small displacement from balance should initially produce growing oscillations. The eigenvalue plot alone does not determine what happens once the solution has moved far from equilibrium.

3.3 What happens after one node is disturbed?

To connect the local calculation with observable trajectories, we started all five nodes at balance and changed only Wood. The first experiment increased Wood by 5%, testing the response near equilibrium. The second

reduced Wood to 30% of its balanced value, testing a much larger initial displacement. The other four components started at balance in both cases. This distinction matters: local stability addresses sufficiently small perturbations, while the larger disturbance requires a direct trajectory calculation.

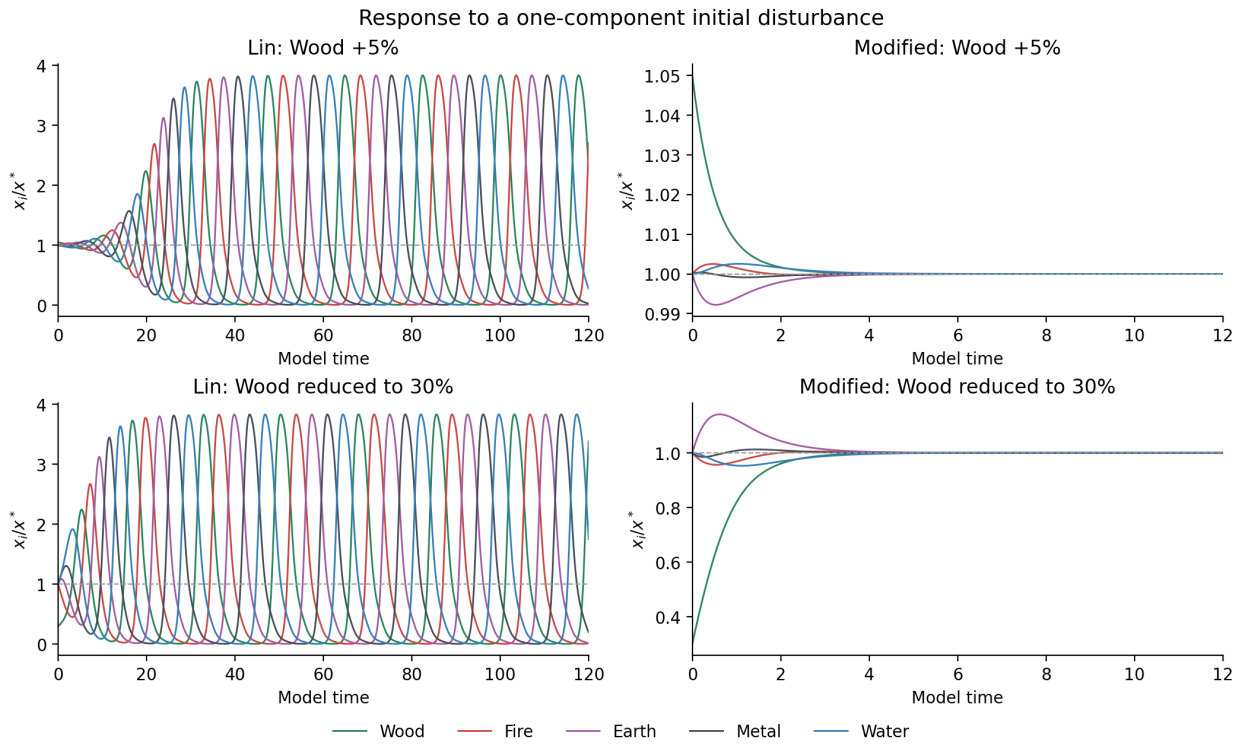


Figure 3: Responses to the same relative initial disturbance at the two baselines in Figure 2. Top: Wood increased by 5%. Bottom: Wood reduced to 30% of balance. Each activity is divided by its model's balanced level, so the horizontal reference value is one. All runs extend to $t = 120$; the modified-model panels display only $t \leq 12$ to show the transient. Axis ranges differ between panels.

In the upper-left panel, the initially small displacement spreads through the network and grows into a sustained oscillatory pattern over the observation interval. This agrees with the local instability identified in Figure 2. The components rise and fall at different times, so the network does not simply move to a new common activity level. The lower-left panel shows that the larger Wood reduction also produces persistent departures from the original balance. At $t = 120$, the Euclidean distances to the balanced vector are approximately 0.8776 and 0.9899 for the small and large disturbances, respectively.

The conclusion is specific: the all-ones Lin baseline does not restore these disturbances to its positive balanced state. Oscillations remain bounded over the displayed interval; their appearance is not evidence of chaos, permanent extinction, or failure of every Lin parameter choice. Indeed, keeping the other coefficients at one but increasing self-limitation to $b = 2$ gives a stable positive balance at $s_{\mathrm{L}} = 0.25$. This observation motivates the parameter sweeps below: a single baseline cannot represent the behavior of the whole model family.

The trajectory graphs were obtained by integrating the full five-node equations with DOP853, relative tolerance 10^{-10} and absolute tolerance 10^{-12}, and sampling 3,001 times over $[0, 120]$. Repeating the four runs with RK45 at the same tolerances changed the sampled activities by at most 1.1×10^{-8}. These checks support the displayed numerical behavior. The present experiments concern the full network; a comparison of isolated or externally constrained two-node motifs remains unfinished.

4 A modified version of the model and its dynamics

4.1 Why examine a different interaction law?

The baseline results raise a modeling question: does changing how a node receives input and experiences losses alter its ability to return to a positive balance? In Lin's model, the nourishment term is proportional to both the mother and the receiving node. Holding the mother fixed, this input becomes smaller as the receiving node becomes weaker, and vanishes when that node is empty. One alternative is to make input available even to a depleted receiver, while reducing it as the receiver's own activity increases. The modified model used in the following experiments is

$$\dot{x}_i = ax_i - kx_i^2 + \frac{cx_{i-1}}{1+x_i} - \frac{gx_i}{1+x_{i+2}} - hx_ix_{i-2}. \tag{3}$$

The input $cx_{i-1}/(1+x_i)$ implements this dependence on receiver activity. The target-dependent loss also changes: holding x_i fixed, a larger target reduces the magnitude of $gx_i/(1+x_{i+2})$. Received restraint remains bilinear. These are mathematical modeling choices whose effects can be explored; the experiments do not establish them as physiological laws.

There are two important qualifications before comparing the plots. First, the direct child-loss term has also been omitted, so the modification changes several mechanisms simultaneously. Second, coefficient names are not fully shared between the models: received restraint is d in Lin and h here. Accordingly, the comparison concerns two complete models rather than one isolated change in functional form. For nonnegative initial activities and positive coefficients, equation (3) preserves nonnegativity: at $x_i = 0$ its derivative is $cx_{i-1} \geq 0$. Thus replenishing a depleted node is compatible with the chosen nonnegative activity domain.

4.2 Does the modified baseline return to balance?

We used $(a, k, c, g, h) = (2, 0.5, 1, 0.5, 0.3)$ as an illustrative baseline. Its positive balanced level is $s_{\mathrm{D}} = (3+\sqrt{59})/4 \approx 2.670286$. The blue markers in Figure 2 all lie in the left half-plane; the largest real part is approximately -1.153724. This predicts decay of sufficiently small disturbances near this balance.

The right-hand panels of Figure 3 show the corresponding network response. After Wood is increased, its excess decays while the other nodes undergo smaller temporary displacements. After Wood is reduced, it rises toward its original level, while some other nodes temporarily overshoot or undershoot theirs. Recovery therefore does not mean that every component moves monotonically or remains unaffected: activity is redistributed transiently before the network returns to the common balance.

Both tested trajectories return close to the starting equilibrium, with final distances of approximately 3.3×10^{-12} and 1.2×10^{-11} at $t = 120$. The small-disturbance experiment is consistent with the eigenvalue prediction. The larger-disturbance experiment adds evidence that recovery is not restricted to the particular 5% test, but it does not determine the full basin of attraction or establish global convergence. The different time and vertical scales in the panels must also be kept in view: the figure demonstrates different outcomes at the chosen baselines, not a calibrated physiological recovery-speed ratio.

4.3 Does the contrast persist when parameters change?

The baseline comparison could depend on the particular coefficients chosen. To move beyond single examples, we varied nourishment and received restraint while holding the remaining coefficients fixed. These axes explore the balance between one positive and one negative contribution to a node's activity. They do not exhaust the models' parameter spaces.

For each of the 90×90 equally spaced grid points, we first found the positive balanced equilibrium, if one existed. For the modified model, this is the larger positive root of $(k+h)s^2 + (k+h-a)s - (a+c-g) = 0$; Lin uses Eq. (2). We then evaluated the Jacobian at that state, computed all five eigenvalues, and colored the point by the largest real part, α. This produces a local stability map, rather than a map obtained by guessing the eventual behavior of a simulated trajectory. Values below -10^{-9} were counted as stable, values above 10^{-9} as unstable, and values within that tolerance as unresolved. The analytic Jacobians were independently checked by numerical differentiation before constructing the plots.

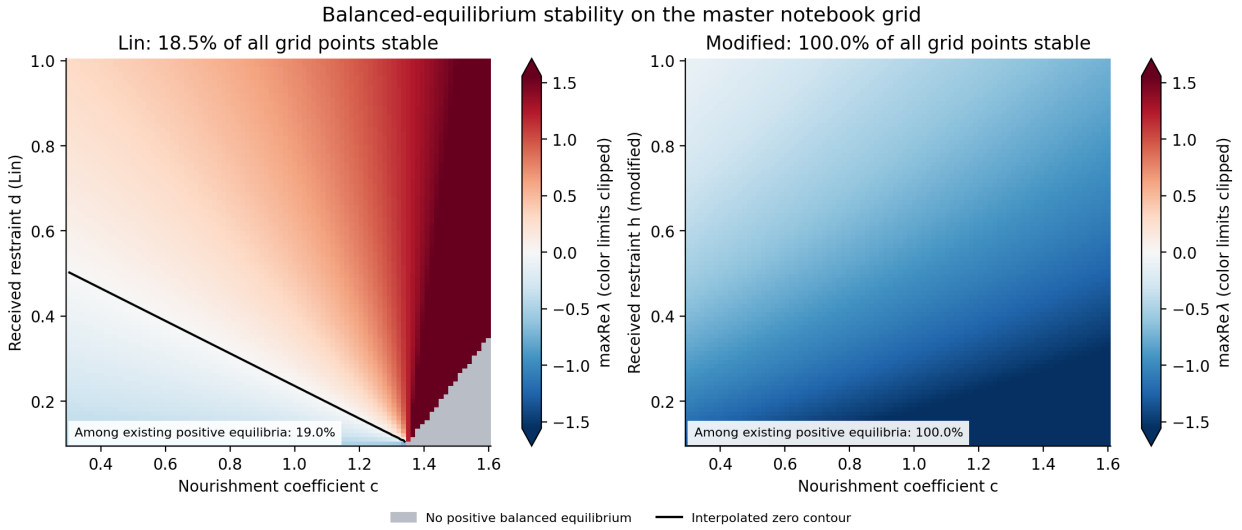


Figure 4: Local stability over $c \in [0.3, 1.6]$ and received restraint $r \in [0.1, 1.0]$. Lin uses $(a, b, c, d, g, h) = (1.5, 0.65, c, r, 0.3, 0.3)$; the modified model uses $(a, k, c, g, h) = (1.5, 0.65, c, 0.3, r)$. Blue and red indicate negative and positive α, respectively; color limits are clipped. Gray marks absence of a positive balanced equilibrium. The black line is an interpolated zero contour. Titles use all grid points as the denominator; inset percentages use only points admitting positive balance.

The Lin panel contains distinct regimes. The lower-left region admits a stable positive balance. Moving toward larger nourishment or larger received restraint within this particular slice crosses into a region where the balance exists but is unstable. The effect of increasing restraint is therefore not automatically stabilizing: stronger negative coupling can coexist with instability of the coupled network. This is an observation about the specified slice, not a claim that restraint always destabilizes the model.

The gray corner has a different interpretation. Here $b + d + g + h - c = 1.25 + d - c \leq 0$, so there is no finite positive balanced level of the form considered. It must not be counted as another unstable positive equilibrium. Of the 8,100 Lin grid points, 7,867 admit positive balance and 1,497 are stable. Hence the two reported percentages are 18.5% of all grid points and 19.0% of eligible points. Near the gray region, color clipping also means that the darkest red does not show the exact magnitude of every positive growth rate.

The modified panel is entirely blue: every grid point in this window admits a locally stable positive balanced equilibrium. This extends the observed contrast beyond the single baseline, but only within this parameter slice. It does not mean that the modified model is stable at every parameter choice. An earlier, different sweep with $a = 2$, $k = 0.5$, $g = 0.5$, $c \in [0.05, 2.5]$, and $h \in [0.05, 2.0]$ had 5,837 stable points out of 8,100 (72.1%). Thus the size of the reported stable region depends on both the window and the fixed coefficients. The appropriate interpretation of Figure 4 is a broader stable region for the modified model in this specified experiment, not a universal advantage.

4.4 How sensitive is the comparison to parameter selection?

A two-dimensional map still fixes several coefficients. We therefore also examined randomly selected parameter sets, asking whether the qualitative contrast persisted under more than one selection rule. The uniform, ordered, and tiered scenarios sample different parts of parameter space; they are sensitivity experiments, not estimates of how frequently the corresponding conditions occur in people.

Each scenario uses 1,200 accepted draws and NumPy's default random-number generator with seed 3. We paired the models through a shared draw (S, A, C, R, T, L) for self-growth, self-limitation, nourishment, received restraint, target-related loss, and child loss. The assignments were $(a, b, c, d, g, h)_{\mathrm{L}} = (S, A, C, R, L, T)$ and $(a, k, c, g, h)_{\mathrm{D}} = (S, A, C, T, R)$. Uniform draws are independent on $[0.1, 2]$. Ordered draws use the same ranges conditioned on $S > C > R > T$, leaving A and L unordered. Tiered draws are independent on $S \in [1, 2]$, $A \in [0.3, 1]$, $C \in [0.5, 1.5]$, $R \in [0.2, 0.8]$, and $T, L \in [0.1, 0.5]$. Using a shared draw makes the assignment explicit, but does not match the strengths of different interaction laws or supply the child-loss term missing from the modified model.

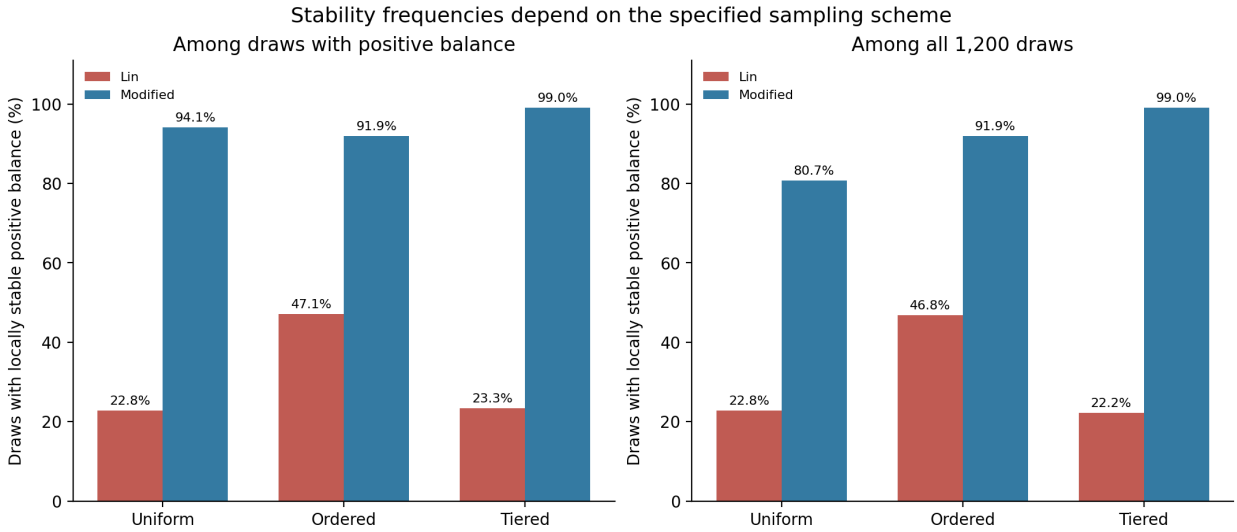


Figure 5: Stability frequencies for the three specified parameter-selection schemes. Left: stable positive balance among draws admitting the selected positive equilibrium. Right: stable positive balance among all 1,200 draws. The paired panels distinguish stability conditional on existence from the combined requirement of existence and stability. Exact counts are listed in Table 1.

The modified model has a higher stability frequency in all three scenarios, but the size of the difference depends on the sampling rule. For example, Lin's conditional frequency rises from 22.8% under uniform sampling to 47.1% under the ordered scenario. Restricting the parameters therefore changes the apparent comparison substantially; the percentages are not fixed characteristics of either model.

The two panels also answer different questions. Under uniform sampling, 94.1% of the modified model's eligible equilibria are stable, but only 80.7% of all draws admit such a stable positive balance. The difference arises because 171 of the 1,200 draws have no positive balanced equilibrium. The left panel asks whether a positive balance is stable once it exists; the right asks how often the chosen sampling procedure yields a stable positive balance at all. Reporting only the left panel would conceal this loss of eligibility. Similarly, the tiered result of 99.0% describes the selected coefficient ranges, not near-certainty of stability under arbitrary parameters.

Scenario	Model	Stable	Positive balance	Total draws
Uniform	Lin	273	1195	1200
	Modified	968	1029	1200
Ordered	Lin	562	1192	1200
	Modified	1103	1200	1200
Tiered	Lin	267	1145	1200
	Modified	1188	1200	1200

Table 1: Counts underlying Figure 5. For the modified model, eligibility and stability refer to the selected upper positive balanced branch. The two denominators are retained explicitly.

4.5 What these comparisons establish

Together, the four graphs address progressively broader questions: whether balance resists small disturbances, how the full network responds to a finite displacement, whether the outcome persists across a parameter slice, and how it changes under different parameter-selection rules. Within the reported experiments, the modified model admits stable positive balance more frequently and recovers from both disturbances at its selected baseline. Lin also admits stable balance, and the modified model is not stable for every parameter choice.

The observations motivate further comparison; they do not identify one cause for the difference. The

models change several interaction terms, and the illustrative baselines also use different coefficients. Removing Lin's child-loss term and then comparing the remaining bilinear model with the modified model would separate that removal from the changes in interaction laws. Further comparisons would be needed to distinguish those laws from one another. Those controlled experiments are separate from the original Lin comparison reported here. The present figures should therefore be read as preliminary evidence about the behavior of two complete models under stated assumptions. Analysis of the associated two-node motifs remains to be completed.

5 Conclusion and Discussion

6 Acknowledgments

References

[1] H. Lin and J. Han. Analysis of Dynamic Five-Element Model. In *Proceedings of the 39th Chinese Control Conference*, pages 777–781, 2020. doi:10.23919/CCC50068.2020.9188438.